

KiCad Tool Training Part 1/2



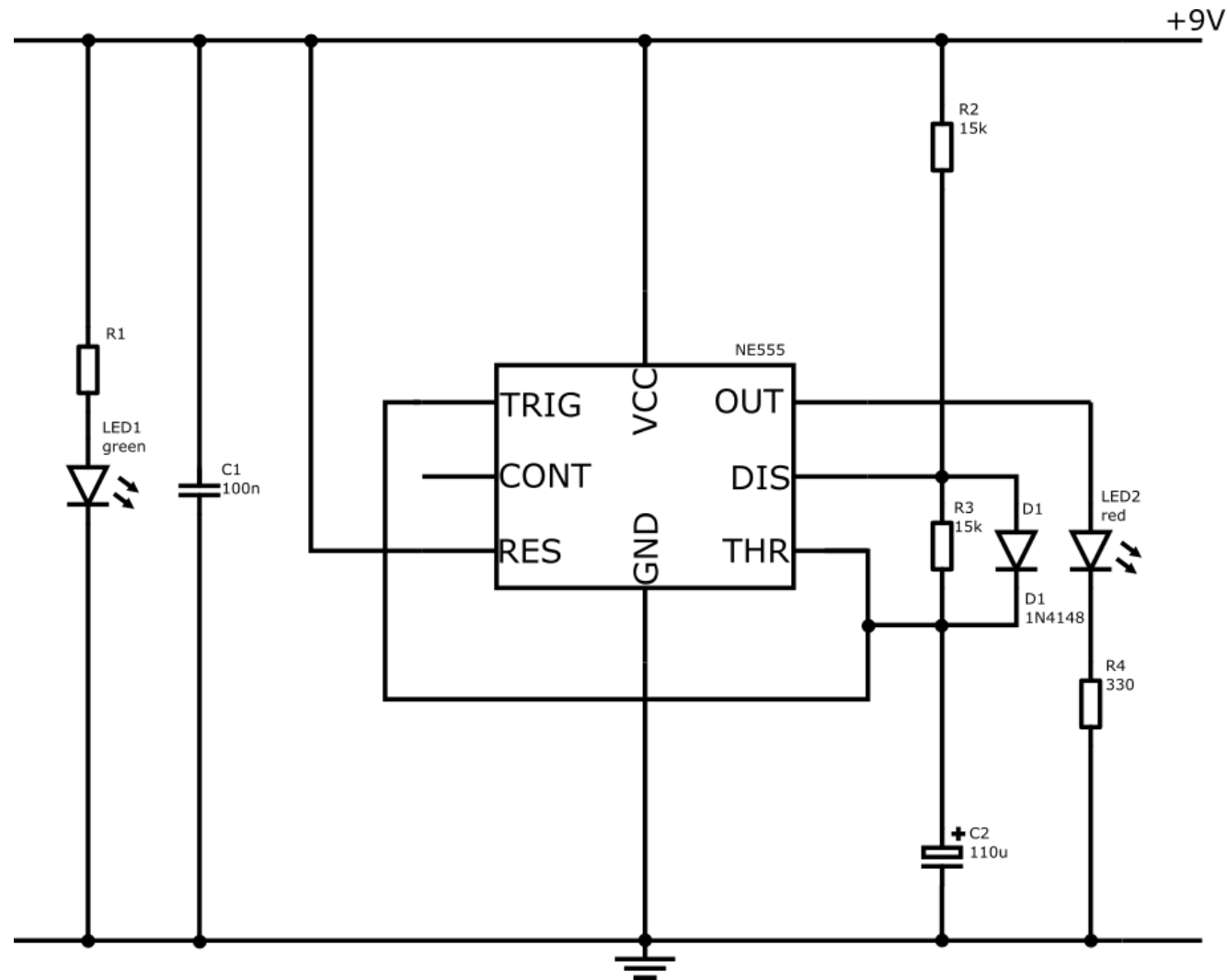
Kicad Workflow & Schematic Editor

KiCad Version 8.0

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University of Applied Science Darmstadt

March 2024



Agenda

- o Installation of KiCad and used Resources
- o KiCad Workflow
- o Schematic Editor

Installing KiCad

<https://kicad-pcb.org/>

Resources for this Training

1. „Getting Started in KiCad“ for Version 8.0, March 2024
2. DigiKeys Videos

Additional Resources for Spring Master Preclass at Hochschule Darmstadt (hda)

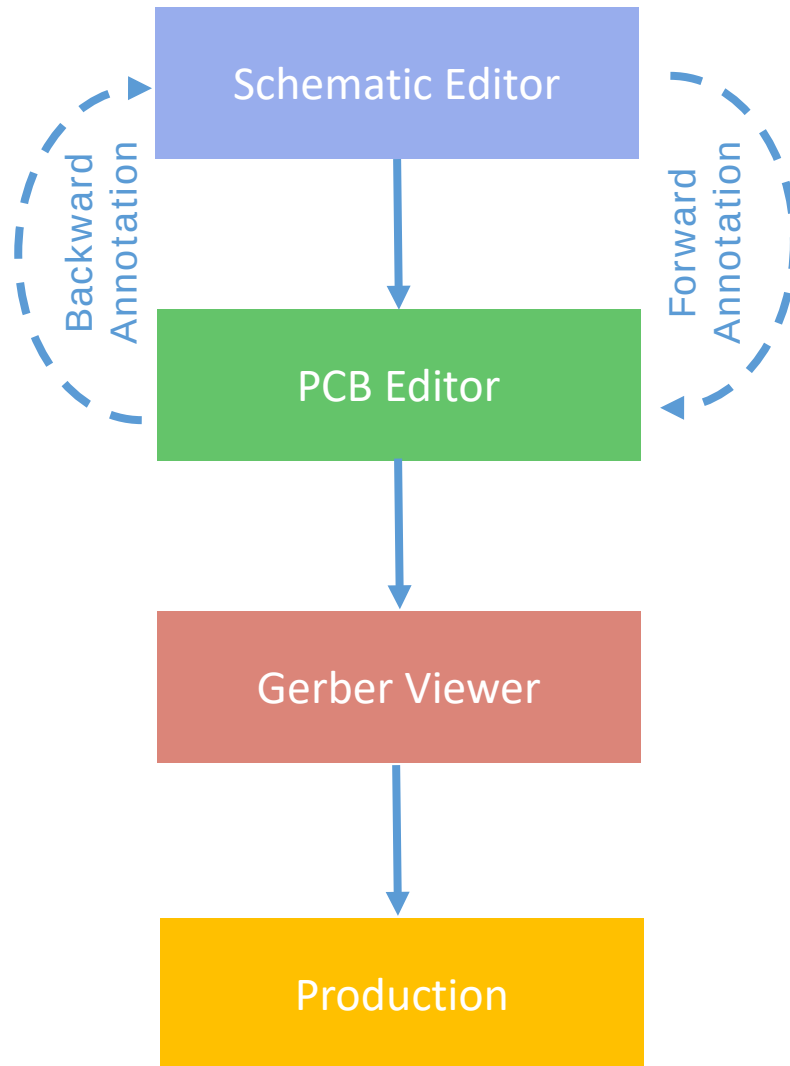
1. KiCad_Training_year.zip
2. KiCad Cheat Sheets

Further Reading

www.555-timer-circuits.com

Agenda

- o Installing KiCad and used Resources
- o KiCad Workflow
- o Schematic Editor

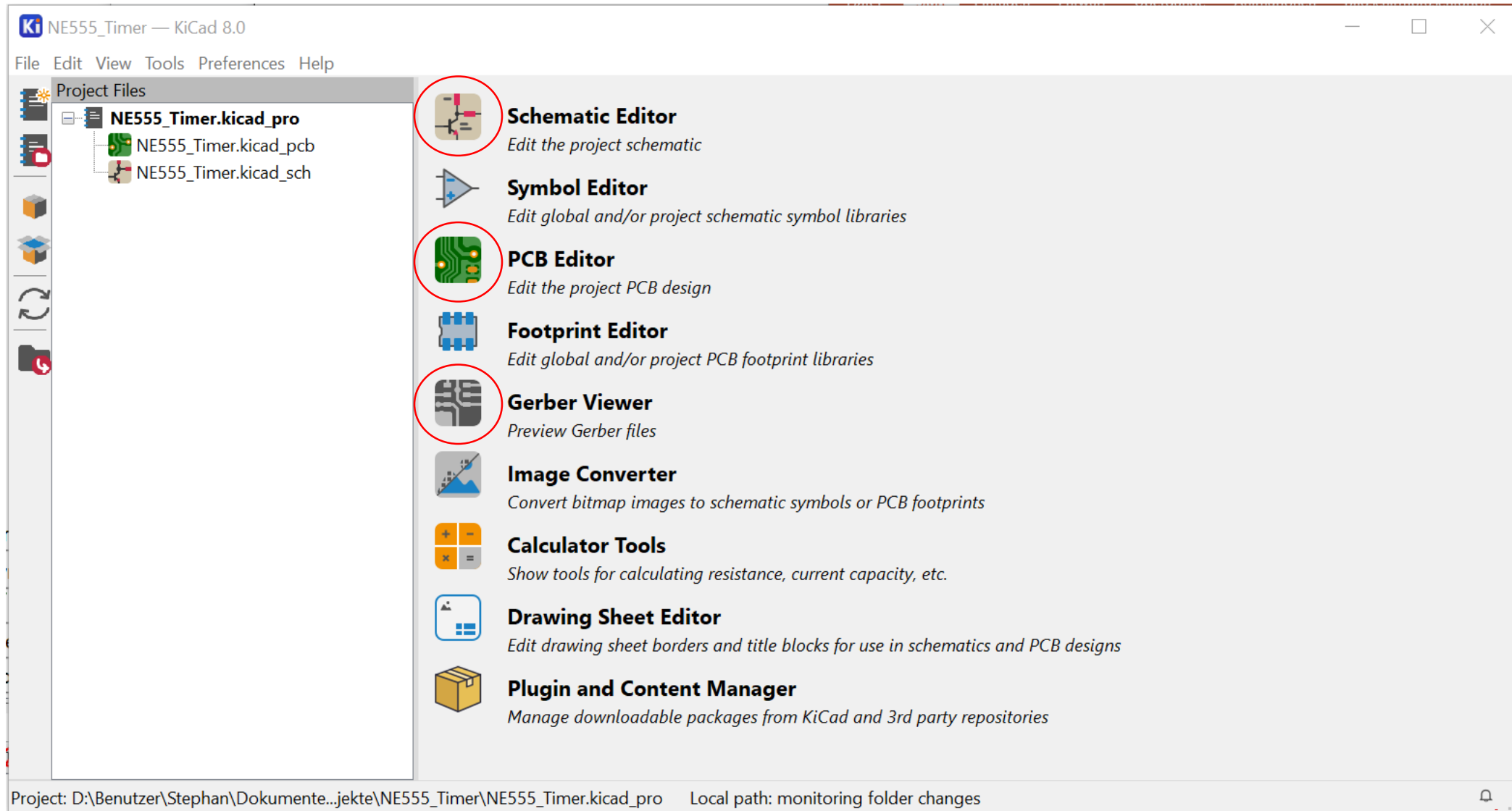


Enter your schematic
Use the schematic component Library

Layout your PCB.
Use the PCB footprint Library

Check your Gerberfiles and send them to
production

Your PCB is produced



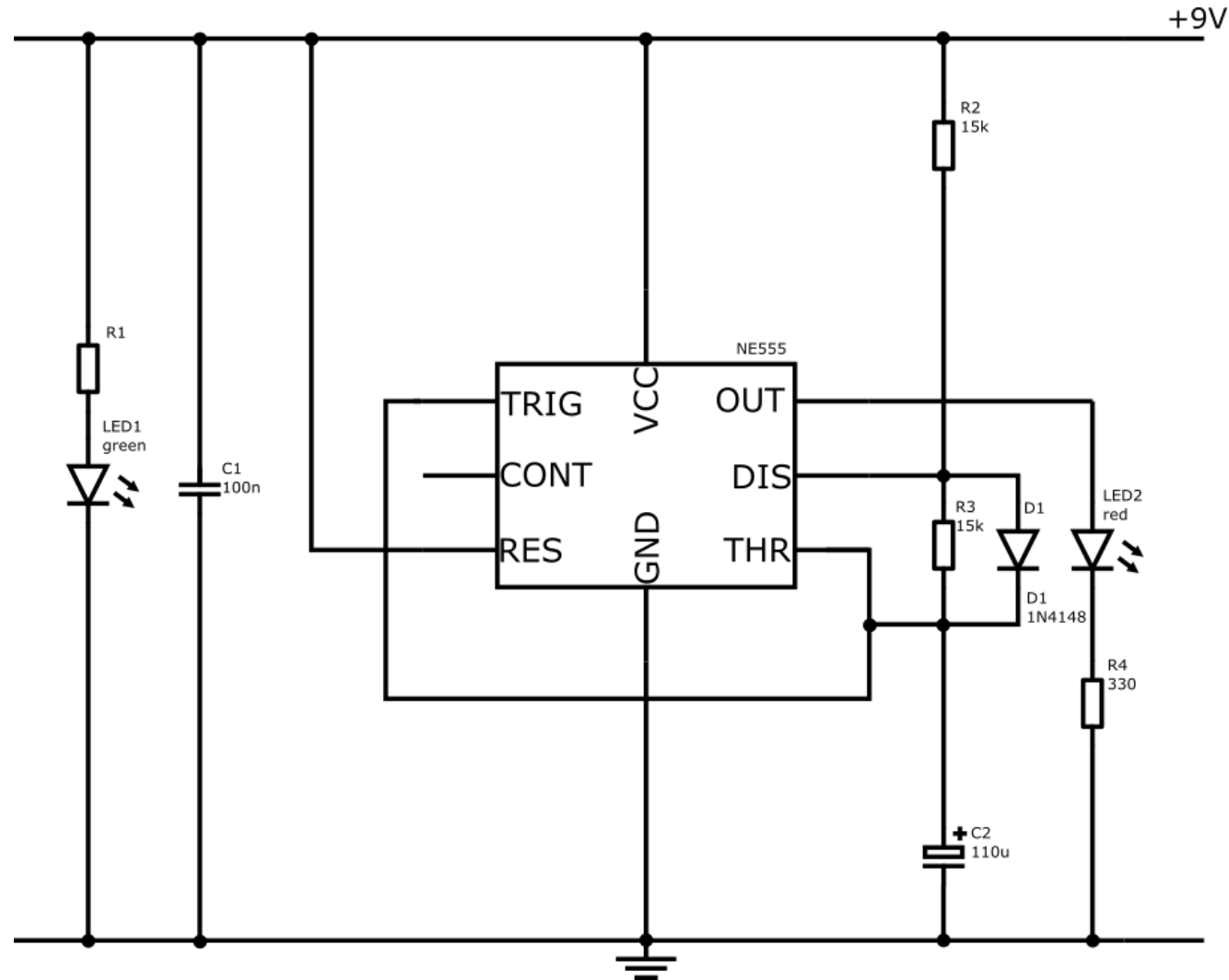
Agenda

- o Installation of KiCad and used Resources
- o KiCad Workflow
- o Schematic Editor



Schematic Entry

1. How to read and draw a schematic
2. Draw a schematic in the Schematic Editor
3. Perform an electrical rule check (ERC)
4. Not Connected Pins and Power Flags
5. Assign Footprints
6. BOM

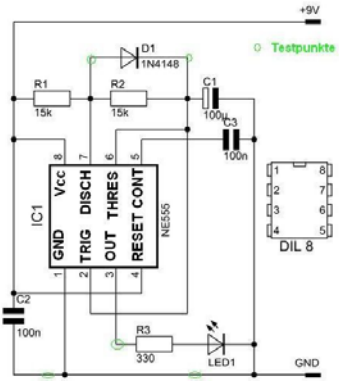




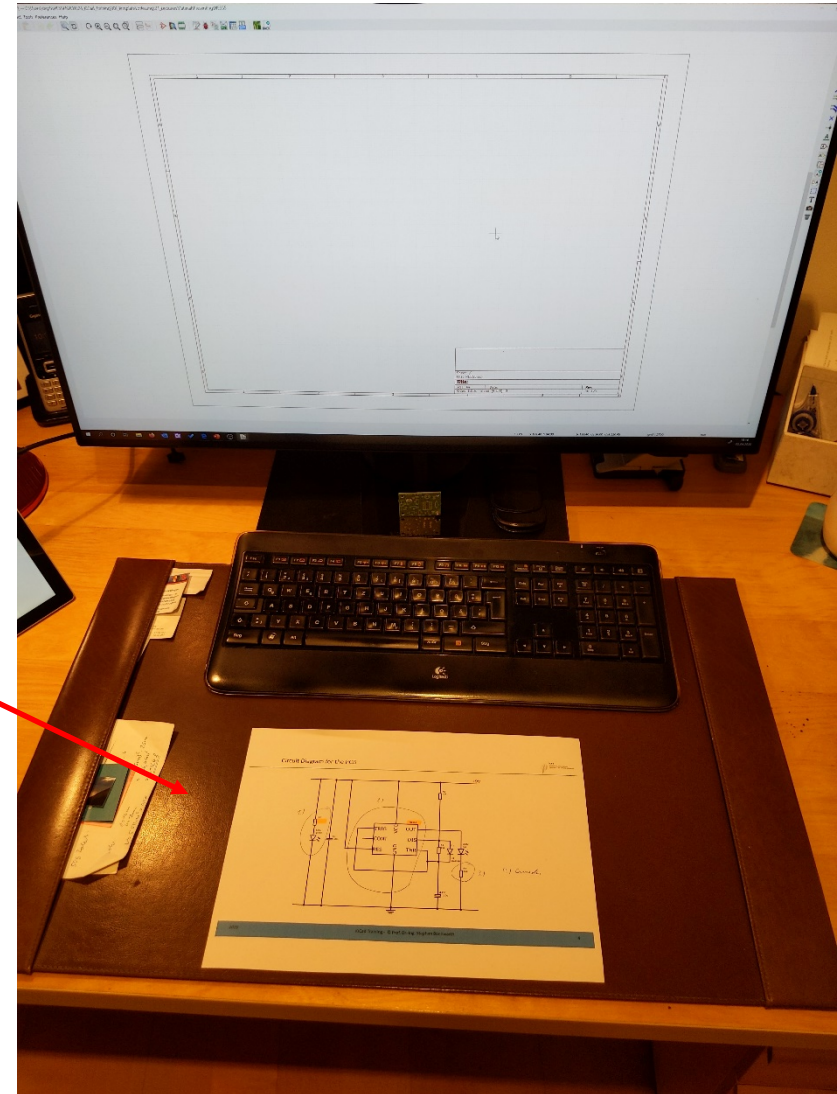
1. How to read and draw a schematic



Put the printed-out
schematic in front of
your keyboard

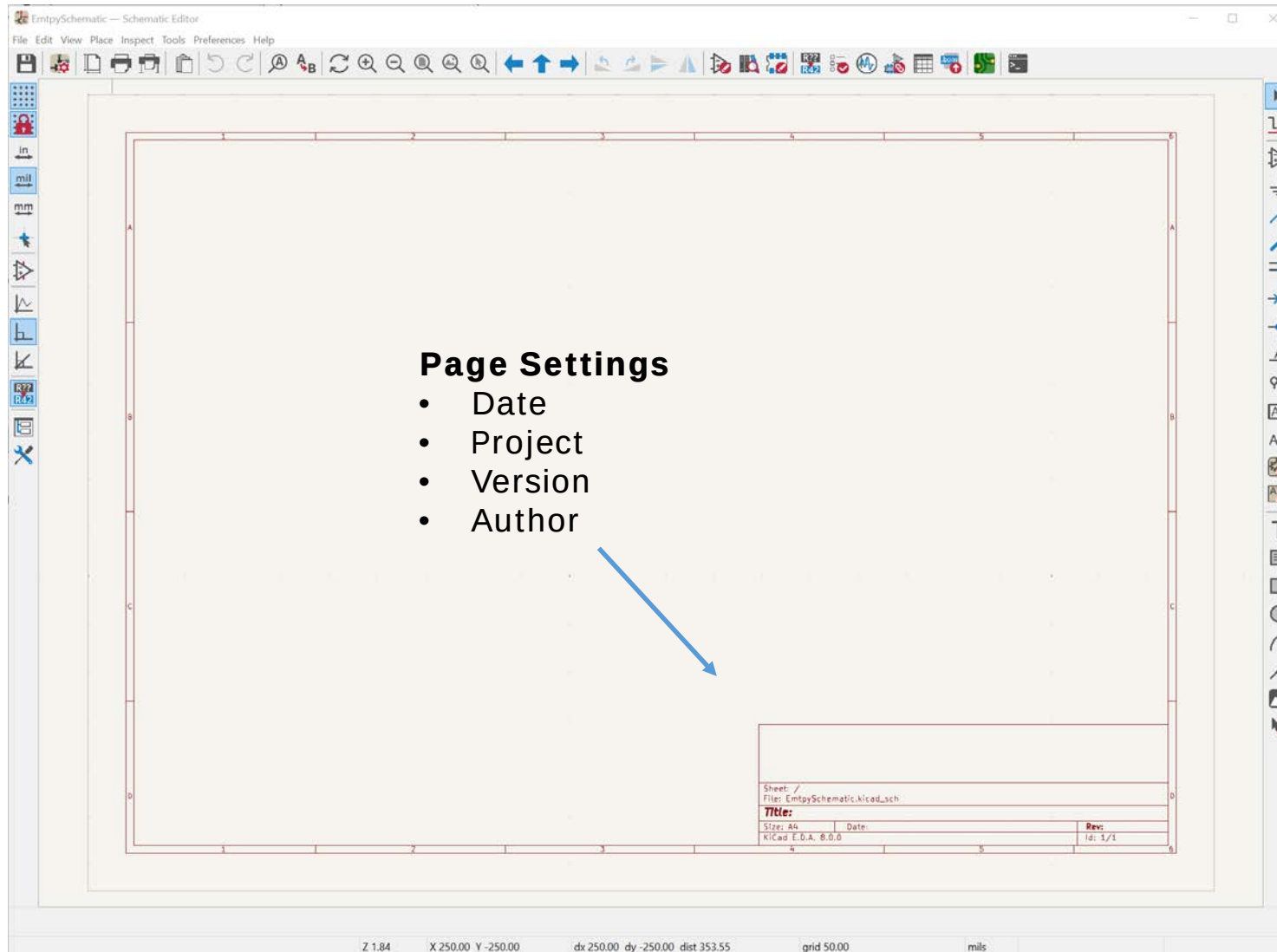


**Engineering
Best
Practice**





1. How to read and draw a schematic



Note on the Schematic Sheet

1. Always fill in the page settings first:

- Later on the PCB Silk Layer
- Date
 - Project
 - Version
 - Lab Group & Student Numbers (Author)

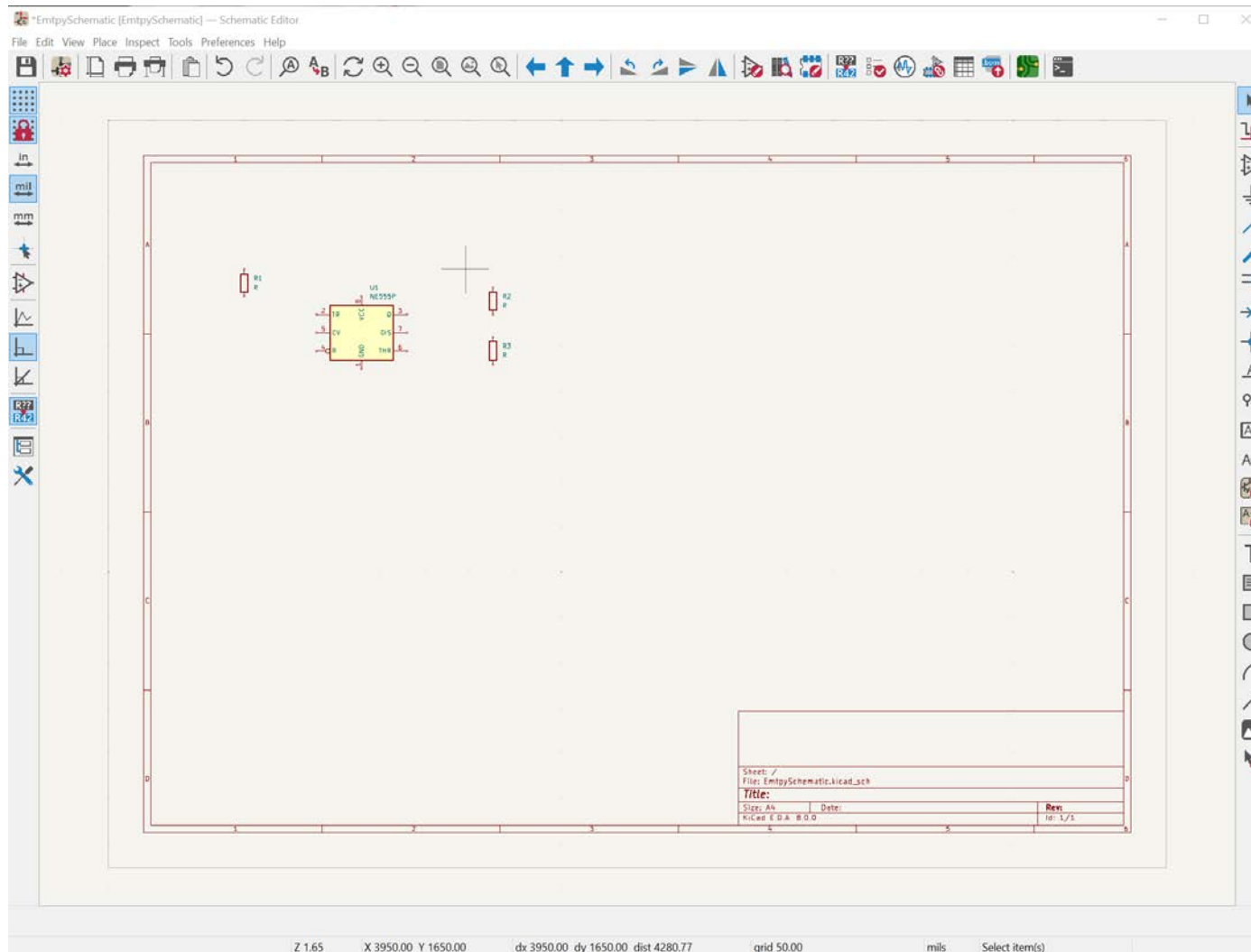
2. Use the sector information A1, C3, ... in your documentation to locate subcircuits.

How to read and draw a schematic

- Left to right -> Input to Output
- Top: Positive Supply
- Bottom: Negative Supply / GND
- Upper right Corner: Supply Voltages
- Center: the circuit performing the task
- Annotations for routing: directly where you have to pay attention
- Text: must be readable in landscape orientation and 90° turned mathematically positive.



2. Draw a schematic in the Schematic Editor



Scrolling and Zooming

Zoom: wheel
Scroll vertically: shift + wheel
Scroll horizontal: ctrl + wheel

Moving

Right click and hold

Working Steps

1. Place all components
 - Supply A1-A2
 - Designed Circuit B1-C3
 - (Auxillary circuits below)
2. Place the connectors
3. Add component values
4. Place the power and ground symbols
5. Wire all up
6. Label nets

a /  → device

v / right click -> prop

a /  → power

w / 

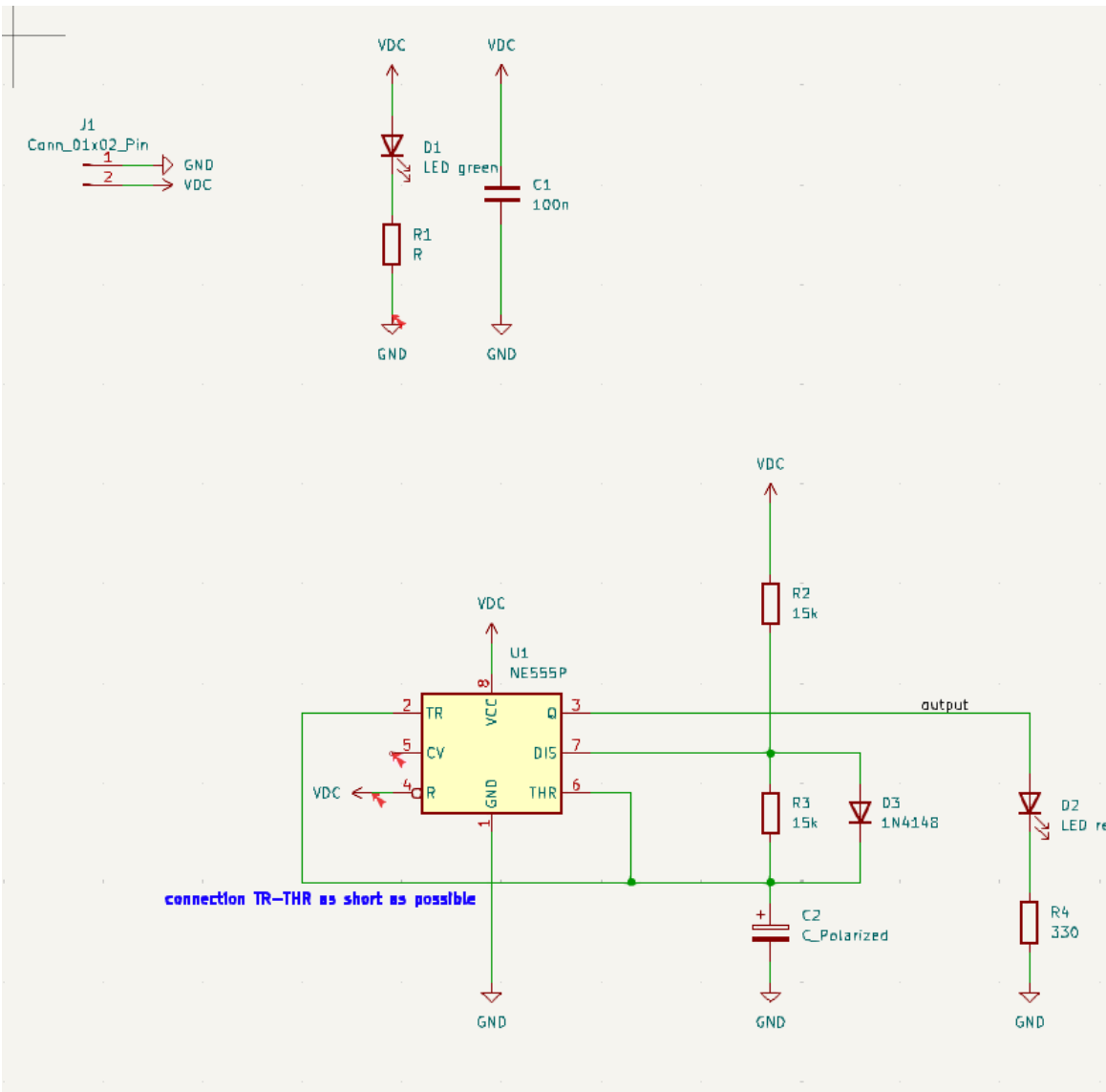
l / 



3. Perform an electrical rule check (ERC)

Working Steps

1. Electrical Rule Check.
Any missing pins or drivers



Electrical Rules Checker

ERC Result

Violations (4) Ignored Tests (1)

- ▼ Error: Input Power pin not driven by any Output Power pins
Symbol #PWR07 Pin 1 [Power input, Line]
- ▼ Error: Pin not connected
Symbol U1 Pin 5 [CV, Input, Line]
- ▼ Error: Input pin not driven by any Output pins
Symbol U1 Pin 5 [CV, Input, Line]
- ▼ Error: Input Power pin not driven by any Output Power pins
Symbol #PWR012 Pin 1 [Power input, Line]

Show: ☐ All

☒ Errors 4

☒ Warnings 0

☐ Exclusions

Save...

Delete Marker

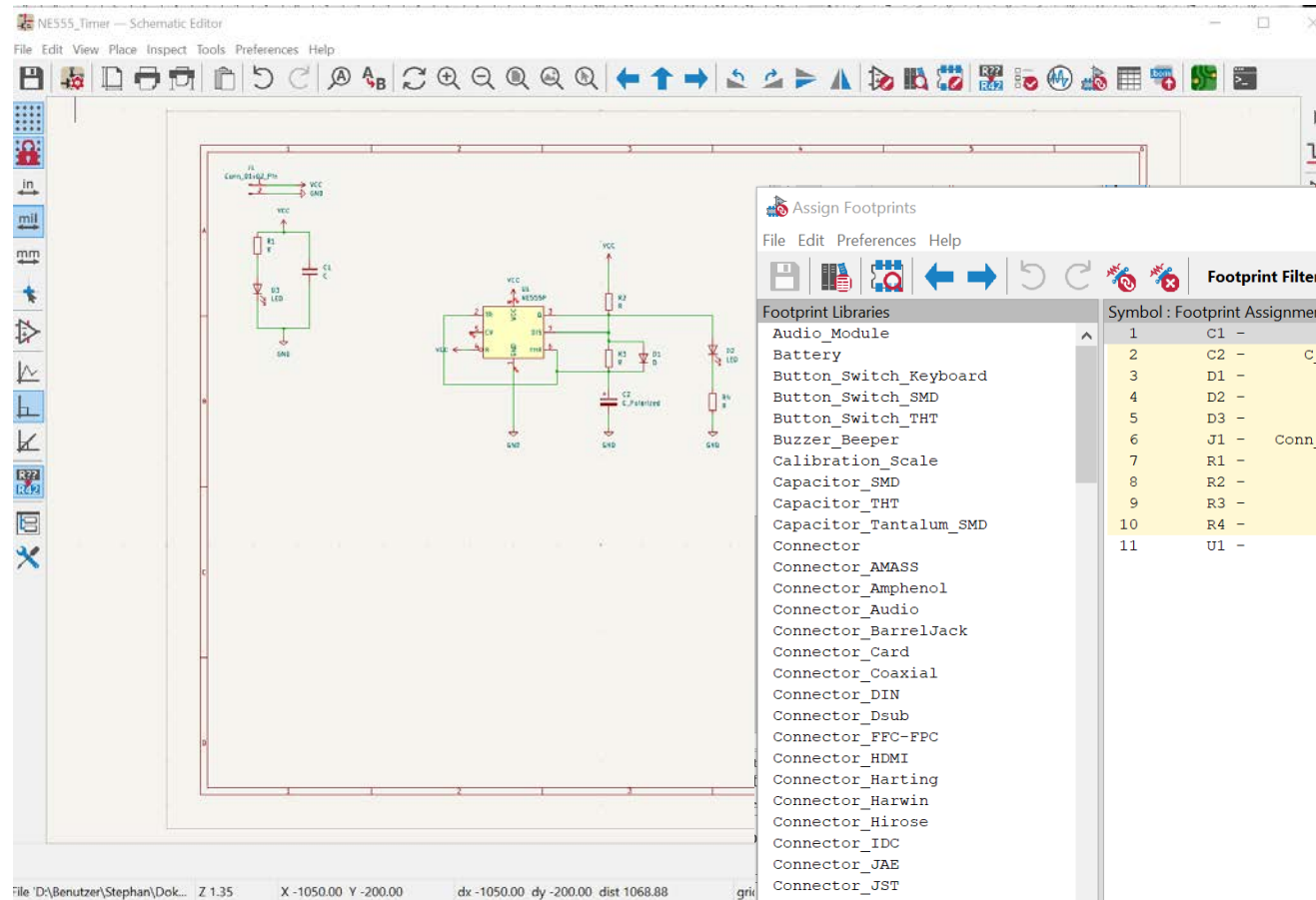
Delete All Markers

Run ERC

Close



5. Assign Footprints



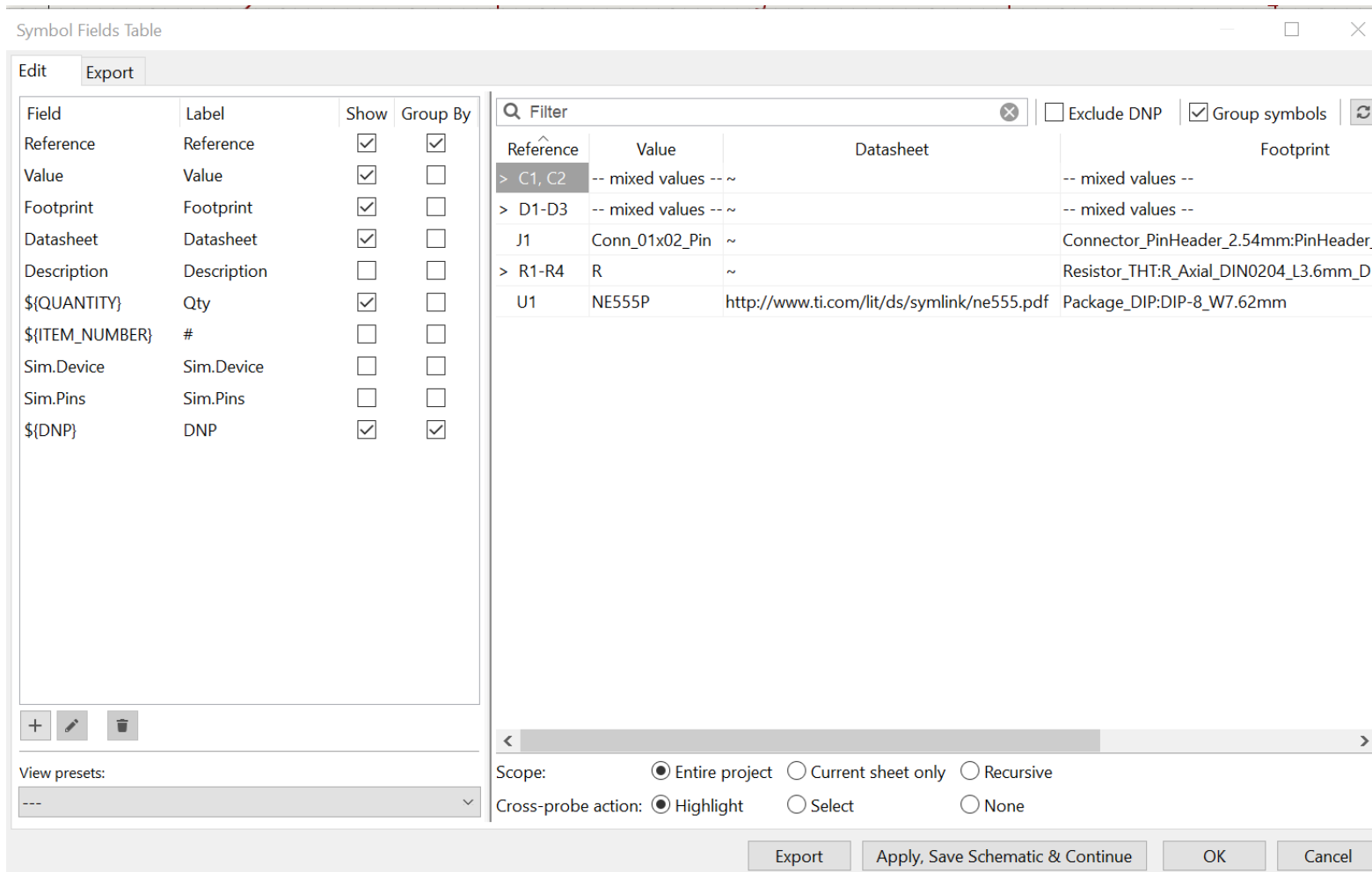
Working Steps

1. Assign Footprints





6. Generate your BOM



Working Steps

1. Generate the Build of Materials (BOM)

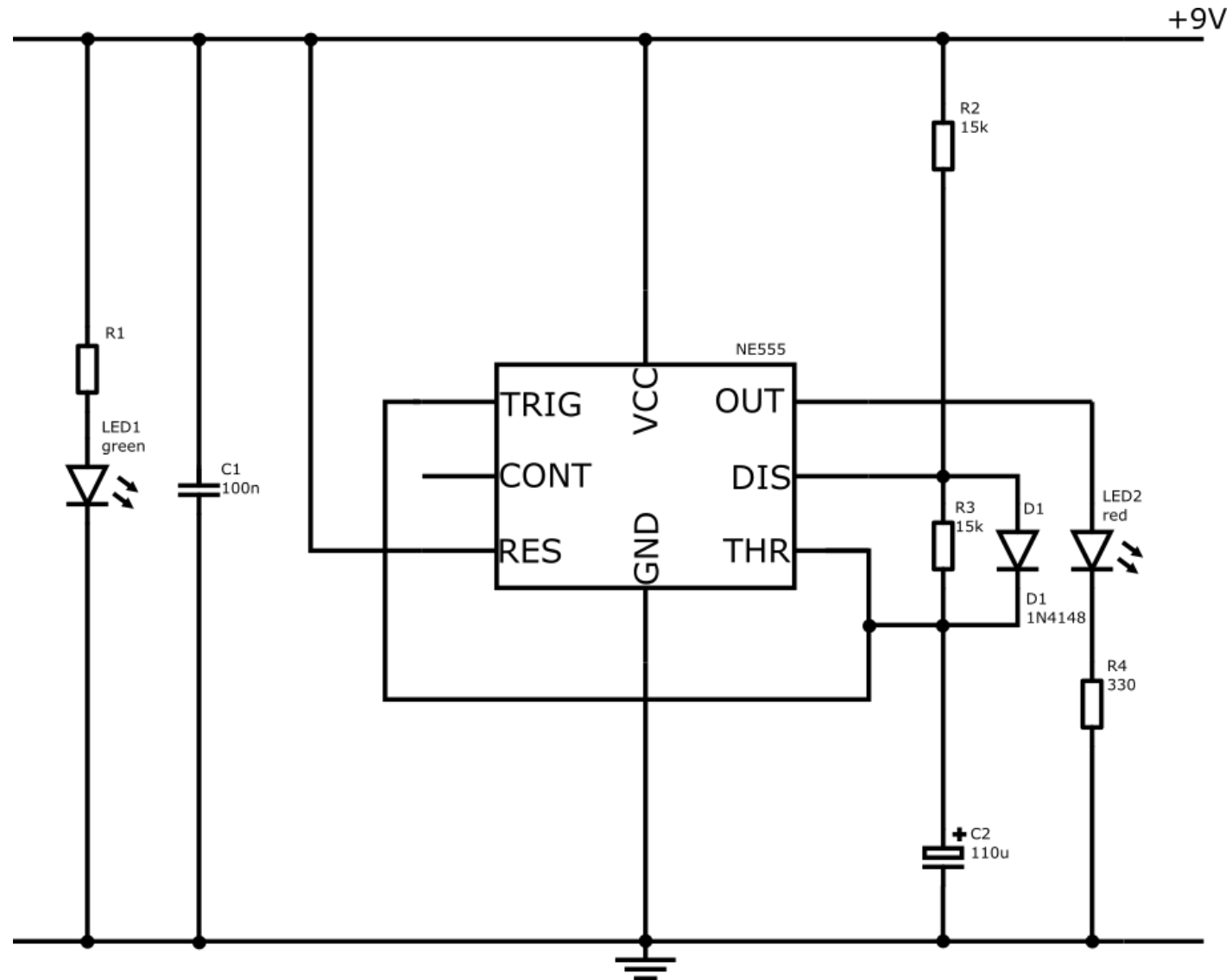


Note on the BOM

1. The components should be grouped by values. This makes it easier to account how many 15k resistances do I have to order?
2. Make a copy of the generate BOM. Manually add mechanical parts, wires, etc.



Exercise: Enter this schematic into the Schematic Editor





Engineering Best Practice

1. Put the schematic **print out** or freehand sketch in front of your keyboard
2. Fill in the project data in the **Worksheet Info Box**
3. The aim is an erc error free schematic. Use the **no connect** and **power flag** symbols.
4. **Understand all measures** and numbers in the footprint description
5. **Add the mechanical parts** to your BOM
6. Order the BOM **in the moment you hand in** your gerber files to the manufacturer.

KiCad Tool Training Part 2/2



Outlook
2nd part of the Tutorial

Layout Editor and Gerber Viewer

Outlook
2nd part of the Tutorial

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